SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Problem-Solution Fit

Date	03 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS]
- Bank Credit Analysts
- Financial Compliance Officers
- Prospective credit card applicants
- Data Scientists in financial institutions

6. CUSTOMER LIMITATIONS [CL]
- Budget: Free tools only (Python, Flask, Kaggle)
- Devices: Web browser only
- Skills: Analysts need simple HTML interface
- Data: Historical repayment data only
- IBM Watson free-tier deployment limits

5. AVAILABLE SOLUTIONS — PROS & CONS [AS]
- Manual review: human judgment but slow & biased
- Credit bureau APIs: trusted but costly
- Generic ML tools: powerful but not tailored
- Our solution: automated, fast, accurate, free

2. PROBLEMS / PAINS + FREQUENCY [PR]
- Manual review is slow and error-prone [Daily]
- No self-service eligibility check [Every application]
- Compliance batch-screening is manual [Weekly]
- Human bias causes inconsistent outcomes [Ongoing]

9. PROBLEM ROOT / CAUSE [RC]
- Banks lack automated ML screening systems
- STATUS codes (0-5, C, X) not usable as labels
- 88%/12% class imbalance makes rules fail
- Risk needs non-linear ML to detect patterns

7. BEHAVIOR + ITS INTENSITY [BE]
- Analysts review applications manually [High-Daily]
- Customers apply without knowing eligibility [Med]
- Officers use Excel for batch analysis [High-Weekly]
- Engineers prepare features manually [High]

3. TRIGGERS TO ACT [TR]
- 500+ credit card applications arrive daily
- Compliance audit needs instant risk identification
- Customers rejected repeatedly without explanation
- Banks want to digitize the approval process

10. YOUR SOLUTION [SL]
- 36,457 applicants merged with binary risk label
- XGBoost model: 78.9% accuracy, 62% recall
- Flask web app: instant single prediction + risk %
- Batch CSV screening for compliance officers
- IBM Watson ML cloud deployment pipeline

8. CHANNELS OF BEHAVIOR [CH]
- ONLINE:
- Flask web app (http://127.0.0.1:5000)
- IBM Watson ML REST API endpoint
- OFFLINE:
- Bank branch manual application forms
- Excel-based compliance screening sheets

4. EMOTIONS BEFORE / AFTER [EM]
- BEFORE — Analysts: overwhelmed by volume
- BEFORE — Customers: anxious after rejection
- BEFORE — Officers: frustrated with manual work
- AFTER — Analysts: confident with ML predictions
- AFTER — Customers: empowered by eligibility check
- AFTER — Officers: efficient with batch screening